

3. To Apply Escape Sequences, Store Command History, Navigate Previous Commands, Navigate Next Commands, Update Input Buffer, Test Recall Functionality.

Code:

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <termios.h>
#include <unistd.h>

#define MAX_INPUT 100
#define MAX_HISTORY 20

char history[MAX_HISTORY][MAX_INPUT];
int history_count = 0;
int history_position = 0;

/* Enable character-by-character input */
void enableRawMode(struct termios *oldSettings) {
    struct termios newSettings;

    tcgetattr(STDIN_FILENO, oldSettings);
    newSettings = *oldSettings;

    newSettings.c_lflag &= ~(ICANON | ECHO);

    tcsetattr(STDIN_FILENO, TCSANOW, &newSettings);
}

/* Restore normal terminal mode */
void disableRawMode(struct termios *oldSettings) {
    tcsetattr(STDIN_FILENO, TCSANOW, oldSettings);
}

/* Store command in history */
void addHistory(const char *command) {
    if (strlen(command) == 0)
        return;

    if (history_count < MAX_HISTORY) {
        strcpy(history[history_count], command);
        history_count++;
    }
    else {
        for (int i = 1; i < MAX_HISTORY; i++) {
            strcpy(history[i - 1], history[i]);
        }

        strcpy(history[MAX_HISTORY - 1], command);
    }

    history_position = history_count;
}

/* Clear current input line */
```

```
GNU nano 8.7.1
/* Clear current input line */
void clearLine(int length) {
    for (int i = 0; i < length; i++) {
        printf("\b \b");
    }

    fflush(stdout);
}

/* Display recalled command */
void displayCommand(char *buffer, int *length) {
    clearLine(*length);
    printf("%s", buffer);
    fflush(stdout);
    *length = strlen(buffer);
}

/* Read command with arrow-key history */
int readCommand(char *buffer) {
    int position = 0;
    char ch;
    buffer[0] = '\0';
    while (1) {
        if (read(STDIN_FILENO, &ch, 1) <= 0)
            return 0;

        /* Enter key */
        if (ch == '\n' || ch == '\r') {
            buffer[position] = '\0';
            printf("\n");
            return 1;
        }

        /* Backspace */
        else if (ch == 127 || ch == 8) {
            if (position > 0) {
                position--;
                buffer[position] = '\0';
                printf("\b \b");
            }
        }
    }
}
```

```

/* Backspace */
else if (ch == 127 || ch == 8) {
    if (position > 0) {
        position--;
        buffer[position] = '\0';
        printf("\b \b");
        fflush(stdout);
    }
}

/* Escape sequence */
else if (ch == 27) {
    char seq1, seq2;

    if (read(STDIN_FILENO, &seq1, 1) <= 0)
        continue;

    if (read(STDIN_FILENO, &seq2, 1) <= 0)
        continue;

    /* Arrow keys */
    if (seq1 == '[') {
        /* UP ARROW */
        if (seq2 == 'A') {
            if (history_count == 0)
                continue;

            if (history_position > 0)
                history_position--;

            strcpy(buffer, history[history_position]);
            displayCommand(buffer, &position);
        }

        /* DOWN ARROW */
        else if (seq2 == 'B') {
            if (history_count == 0)
                continue;

            if (history_position < history_count - 1) {
                history_position++;
            }
        }
    }
}

```

```

GNU nano 8.7.1
    }

    /* DOWN ARROW */
    else if (seq2 == 'B') {
        if (history_count == 0)
            continue;

        if (history_position < history_count - 1) {
            history_position++;
            strcpy(buffer, history[history_position]);
            displayCommand(buffer, &position);
        }
        else {
            history_position = history_count;
            buffer[0] = '\0';
            displayCommand(buffer, &position);
        }
    }
}

/* Normal character */
else if (position < MAX_INPUT - 1) {
    buffer[position++] = ch;
    buffer[position] = '\0';

    printf("%c", ch);
    fflush(stdout);
}
}

int main() {
    char input[MAX_INPUT];
    struct termios oldSettings;

    printf("Dynamic Shell Input & Command History\n");
    printf("Use UP/DOWN arrows to recall commands.\n");
    printf("Type 'history' to view commands.\n");
    printf("Type 'exit' to quit.\n\n");

    enableRawMode(&oldSettings);
}

```

```

GNU nano 8.7.1
int main() {

    char input[MAX_INPUT];
    struct termios oldSettings;

    printf("Dynamic Shell Input & Command History\n");
    printf("Use UP/DOWN arrows to recall commands.\n");
    printf("Type 'history' to view commands.\n");
    printf("Type 'exit' to quit.\n\n");

    enableRawMode(&oldSettings);

    while (1) {

        printf("shell> ");
        fflush(stdout);

        readCommand(input);

        /* Exit */
        if (strcmp(input, "exit") == 0) {
            break;
        }

        /* Display history */
        if (strcmp(input, "history") == 0) {
            printf("Command History:\n");

            for (int i = 0; i < history_count; i++) {
                printf("%d %s\n", i + 1, history[i]);
            }

            continue;
        }

        /* Store command */
        if (strlen(input) > 0) {
            addHistory(input);
            printf("Command entered: %s\n", input);
        }
    }

    disableRawMode(&oldSettings);
    printf("Memory released successfully.\n");

    return 0;
}

```

Result:

```

saisharan134@APL-5CD1366K5L: ~/
saisharan134@APL-5CD1366K5L:~/SKILL$ nano Skill3a.c
saisharan134@APL-5CD1366K5L:~/SKILL$ gcc Skill3a.c -o S3a
saisharan134@APL-5CD1366K5L:~/SKILL$ ./S3a
Dynamic Shell Input & Command History
Use UP/DOWN arrows to recall commands.
Type 'history' to view commands.
Type 'exit' to quit.

shell> hello
Command entered: hello
shell> ls
Command entered: ls
shell> pwd
Command entered: pwd
shell> test
Command entered: test
shell> history
Command History:
1 hello
2 ls
3 pwd
4 test
shell> exit
Memory released successfully.
saisharan134@APL-5CD1366K5L:~/SKILL$

```

To Allocate Buffers Dynamically, Resize Arrays, Prevent Buffer Overflow, Manage Linked Lists, Release Memory Correctly, Verify with Valgrind.

Code:

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define INITIAL_SIZE 2

/* Linked list node */
typedef struct Node {
    char *command;
    struct Node *next;
} Node;

/* Add a command to linked list */
void addNode(Node **head, const char *command) {
    Node *newNode = malloc(sizeof(Node));

    if (newNode == NULL) {
        printf("Memory allocation failed.\n");
        return;
    }

    newNode->command = malloc(strlen(command) + 1);

    if (newNode->command == NULL) {
        free(newNode);
        printf("Memory allocation failed.\n");
        return;
    }

    strcpy(newNode->command, command);
    newNode->next = NULL;

    if (*head == NULL) {
        *head = newNode;
    }
    else {
        Node *temp = *head;

        while (temp->next != NULL) {
            temp = temp->next;
        }

        temp->next = newNode;
    }
}

/* Display linked list */
void displayList(Node *head) {
    Node *temp = head;

    /* Display linked list */
    void displayList(Node *head) {
        Node *temp = head;

        printf("\nLinked List Commands:\n");

        while (temp != NULL) {
            printf("- %s\n", temp->command);
            temp = temp->next;
        }
    }

    /* Free linked list memory */
    void freeList(Node *head) {
        Node *temp;

        while (head != NULL) {
            temp = head;
            head = head->next;
            free(temp->command);
            free(temp);
        }
    }

    int main() {
        int capacity = INITIAL_SIZE;
        int count = 0;

        /* Dynamic array allocation */
        char **commands = malloc(capacity * sizeof(char *));

        if (commands == NULL) {
            printf("Memory allocation failed.\n");
            return 1;
        }

        Node *head = NULL;

        printf("Dynamic Memory Management Demo\n");
        printf("-----\n");

        /* Commands to store */
        const char *input[] = {
            "ls",
            "pwd",
            "cd Documents",

```

GNU nano 8.7.1

```
/* Commands to store */
const char *input[] = {
    "ls",
    "pwd",
    "cd Documents",
    "echo Hello",
    "gcc program.c"
};

int totalCommands = 5;

for (int i = 0; i < totalCommands; i++) {

    /* Resize array when full */
    if (count == capacity) {

        capacity *= 2;

        char **temp = realloc(
            commands,
            capacity * sizeof(char *)
        );

        if (temp == NULL) {
            printf("Memory reallocation failed.\n");

            /* Free previously allocated memory */
            for (int j = 0; j < count; j++) {
                free(commands[j]);
            }

            free(commands);
            freeList(head);

            return 1;
        }

        commands = temp;

        printf("\nBuffer resized to capacity: %d\n", capacity);
    }

    /* Allocate memory for command */
    commands[count] = malloc(strlen(input[i]) + 1);

    if (commands[count] == NULL) {
        printf("Memory allocation failed.\n");

        for (int j = 0; j < count; j++) {
            free(commands[j]);
        }
    }
}
```

GNU nano 8.7.1

```
/* Allocate memory for command */
commands[count] = malloc(strlen(input[i]) + 1);

if (commands[count] == NULL) {
    printf("Memory allocation failed.\n");

    for (int j = 0; j < count; j++) {
        free(commands[j]);
    }

    free(commands);
    freeList(head);

    return 1;
}

strcpy(commands[count], input[i]);

/* Add command to linked list */
addNode(&head, input[i]);

count++;

printf("Stored command: %s\n", input[i]);
}

/* Display dynamic array */
printf("\nDynamic Array Contents:\n");
for (int i = 0; i < count; i++) {
    printf("%d. %s\n", i + 1, commands[i]);
}

/* Display linked list */
displayList(head);

/* Release dynamic array memory */
for (int i = 0; i < count; i++) {
    free(commands[i]);
}

free(commands);

/* Release linked list memory */
freeList(head);

printf("\nAll dynamically allocated memory released successfully.\n");

return 0;
}
```

Result:

```
saisharan134@APL-5CD1366K5L: ~/  + v
saisharan134@APL-5CD1366K5L:~/SKILL$ nano Skill3b.c
saisharan134@APL-5CD1366K5L:~/SKILL$ gcc Skill3b.c -o S3b
saisharan134@APL-5CD1366K5L:~/SKILL$ ./S3b
Dynamic Memory Management Demo
-----
Stored command: ls
Stored command: pwd

Buffer resized to capacity: 4
Stored command: cd Documents
Stored command: echo Hello

Buffer resized to capacity: 8
Stored command: gcc program.c

Dynamic Array Contents:
1. ls
2. pwd
3. cd Documents
4. echo Hello
5. gcc program.c

Linked List Commands:
- ls
- pwd
- cd Documents
- echo Hello
- gcc program.c

All dynamically allocated memory released successfully.
saisharan134@APL-5CD1366K5L:~/SKILL$
```